

BACHELOR WITH COMPUTER APPLICATIONS AS MAJOR (CT-I)
5th SEMESTER

CAP522J1: COMPUTER APPLICATIONS _ OPERATING SYSTEM

CREDITS: THEORY (3) PRACTICAL (1)

COURSE LEARNING OUTCOMES:

At the end of this course, a student should be able to:

- To learn and understand the concepts of operating system.
- To understand the core structure and functions of operating system.
- Explain process management, processor scheduling, and concurrent programming.
- Understand the concept of deadlocks and synchronization.
- Learn memory management, file management. I/O systems, and disk scheduling.
- Distinguish main memory and virtual memory.
- Learn Operating System Design issues.
- Learn basic Unix commands and shell programming

UNIT-1 (15 HOURS)

Introduction: Goals & Structure of Operating System. Basic functions & Modes. Types of Operating System (Batch, Multiprogramming, Multitasking, Time Sharing, Parallel, Distributed A Real-Time OS). System calls.

Process: Process Concept, Process states, Threads, Uni-processor Scheduling: Types of scheduling (Pre-emptive & Non- pre-emptive). Scheduling algorithms (FCFS, SJF, RR, Priority, Thread Scheduling).

UNIT-2 (15 HOURS)

Concurrency: Principles of Concurrency, Mutual Exclusion: Software approaches, Hardware Support, Semaphores, Message Passing, Signals. Monitors, Classical Problems of Synchronization (Readers-Writers, Producer-Consumer, and Dining Philosopher problem).

Deadlock: Principles of Deadlock. Deadlock Prevention & Avoidance, Deadlock Detection.

UNIT-3 (15 HOURS)

Memory: Memory Management requirements, Memory partitioning (Fixed and Variable Partitioning), Memory Allocation Strategies (First Fit, Best Fit and Worst Fit), Fragmentation. Swapping, Segmentation, Paging.

Virtual Memory: Virtual Memory Concepts and Implementation, Concept of Demand Paging, Page Replacement Policies (FIFO, LRU, Optimal, Other Strategies). Thrashing.

PRACTICALS (I CREDIT: 30 HOURS)

- In Installing and setting up of Linux /Unix Operating system.
- Using Basic Unix commands like echo, printf. Is, who, date, passwd, cal.
- Using pwd, cd, mkdir, rmdir, cat commands
- Using mv, rm, cp, wc and od commands.
- Demonstrate the use of Pipes.
- Using Basic and Extended regular expressions
- Implementing file permissions, printing commands.
- Mount devices like usb drives in UNIX/Linux.
- Write a shell script to demonstrate the use of variable & basic input output on the console.
- Write a script to demonstrate the use of control statements

TEST BOOK:

- Peter B Galvin, Greg Gagne. Abraham Silberschatz: Operating System Concepts. John Wiley & Sons, Inc.
- Shell Programming by Yashwant Kanetkar

REFERENCE BOOKS:

- Peter B Galvin. Greg Gagne. Abraham Silberschatz: Operating System Concepts. John Wiley A Sons, Inc.
- Andrew S. Tanenbaum: Modern Operating Systems, PHI
- William Stallings: Operating Systems, Pearson Education India
- M. G. Venkatesh Murthy: UNIX & Shell Programming, Pearson Education
- Richard Blum, Christine Bresnahan: Linux Command Line and Shell Scripting Bible, Wiley.